

Pearse Jim

Machine Learning Research Engineer

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SUMMARY

Machine learning research engineer working on model efficiency and low-resource languages, with four 2026 papers and the systems behind them. I build the measurement before the model: a harness that decides which agent ships rather than an opinion, speaker-disjoint splits that stop a tone classifier memorising voices, per-channel calibration that recovers reasoning a naive INT4 quantiser destroys. Recent work spans compressing recursive reasoners onto 4 MB microcontrollers, a Yoruba pronunciation-assessment engine trained on ~118 h of diacritised speech, and an SEC 10-K agent harness with 100% provenance integrity.

SKILLS

Languages: Python, C++, SQL, Bash

ML & Research: PyTorch, Transformers, mixture-of-experts and sparse routing, recursive/latent-state reasoners, quantisation (INT8, INT4, per-channel calibration, QAT), structured pruning, distillation, linear attention, scikit-learn, CatBoost, NumPy, pandas

Speech & Audio: Montreal Forced Aligner (Kaldi GMM-HMM), wav2vec2, Meta MMS, Whisper, Praat/Parselmouth, F0 and tone modelling, G2P and pronunciation dictionaries, goodness-of-pronunciation, WER/BLEU/chrF with bootstrap CIs

LLM & Agents: OpenAI, Anthropic, DeepSeek, Google AI, local Ollama; LangGraph, Model Context Protocol, tool-calling agent loops, BM25 and graph retrieval, evaluation harnesses, LLM-as-judge, prompt versioning, regression detection

Serving & Data: FastAPI, Flask, Streamlit, PostgreSQL, SQLite, Docker, background job queues, Render and Fly.io, SEC EDGAR and XBRL, large-scale scraping, Power BI, Tableau

Practice: reproducible evaluation, speaker-disjoint and hand-verified ground truth, stratified reporting over aggregates, ablation design, pytest

PUBLICATIONS

- **A Tilted Bowl Is Not a Slippery Slope: Compressing Looped Models.** Steven Kolawole, *Pearse Jim*, Opegbemi M. Busoye, Glory Bagai, Virginia Smith (ML Collective; Carnegie Mellon University). arXiv:2609.39277, 2026. *Equal contribution.
- **What Survives When You Compress a Recursive Reasoner for the Edge?** Pearse Jim, *Steven Kolawole*, Opegbemi M. Busoye, Glory Bagai, Virginia Smith (ML Collective; Carnegie Mellon University). arXiv:2606.26488, 2026. *Equal contribution.
- **Why Large Language Models Fail for Hausa Educational Content: Cascading Errors from Translation to Speech to Comprehension.** Accepted at WiML @ ICML 2026 and ICBINB @ ICLR 2026. First author.
- **Remix, Don't Expand: Context-Aware Embedding Routing.** Accepted at LM4UC @ AAAI 2026. Sole author.

EXPERIENCE

Independent Machine Learning Researcher | ML Collective | Remote · April 2025 – Present

- First author on edge compression of recursive reasoners: swept precision across three tasks and two recursive architectures and showed that aggressive compression preserves local prediction while destroying global reasoning: cell accuracy holds as puzzle-exact accuracy collapses to zero under naive INT4, pruning, distillation and linear attention alike.
- Introduced **carry-trajectory fidelity**, a label-free cosine signal against the full-precision reasoning path that predicts that collapse and its recovery before any task evaluation is run.
- Turned the finding into a deployment recipe: flash-streamed embeddings remove a 99.4 MB bottleneck (79% of the 126 MB baseline), INT8 at one recursion cycle matches full-depth accuracy at 6× fewer FLOPs and fits an 8 MB SoC, and per-channel calibrated INT4 fits a 4 MB microcontroller without retraining.
- Built the first stratified evaluation of the English→Hausa→speech→comprehension pipeline for exam content: 790 balanced WASSCE questions over 10 subjects and 15 years, five ASR models across three audio branches (human speech and two TTS voices).
- Showed that aggregate WER hides the failures that matter. Subject-level WER differs significantly (Kruskal-Wallis $H=66.0$, $p<1e-6$; Economics 54.6% against Chemistry 75.6%), and 63.4% WER costs 95.6% of downstream translation quality (4.44 BLEU against a perfect-transcript reference), with tonal/long-vowel errors at 100% against 63.9% for short vowels.
- Applied mixture-of-experts routing directly to the embedding table rather than to feed-forward layers: a remixing router over a compact 64-dimensional base matches or beats dense 512-dimensional baselines on five language-modelling benchmarks with 8× fewer active dimensions and 82.5% fewer parameters (12.32M against 70.46M), a 3.6× parameter-efficiency gain on Penn Treebank.

Tech: PyTorch, Transformers, quantisation and QAT, MoE routing, wav2vec2/Whisper/MMS, bootstrap CIs, Kaggle/GPU training.

Software Engineer & Machine Learning Engineer | AoLabs | Remote · October 2024 – April 2025 (Contract)

- Benchmarked a novel weightless neural architecture against Transformer and LSTM baselines across multiple tasks, building the evaluation harness the comparison ran on.
- Built a continually-learning movie recommender using online learning, and reinforcement-learning game agents on the same architecture.
- Built large-scale scraping pipelines feeding model training and evaluation, and deployed the resulting apps.
- Open source: extended the AoLabs MNIST project from binary to greyscale classification.

Tech: Python, PyTorch, Flask, scikit-learn, CatBoost, reinforcement learning.

Machine Learning Engineer Intern, Assistant Project Lead | ACME | March 2024 – September 2024

- Built and deployed an artist similarity and recommendation service on Flask with gradient-boosted models.
- Led a sub-team through data collection, feature engineering, model selection and production deployment.

Machine Learning Intern, Assistant Project Lead | Hamoye | May 2022 – November 2022

- Assistant Project Lead for Team SciPy at the Hamoye '22 Grand Finale: owned scope, timelines and team communication.
- Built and deployed a traffic-prediction application on Streamlit; coordinated the visualisation notebooks with the Project Lead and Query Analyst.

Engineering Intern | SpeedVibe | September 2021 – April 2022

- Developed embedded systems and IoT devices; wrote firmware in C++ on Arduino, with component selection, evaluation and technical documentation.

SELECTED PROJECTS

caseflow: Customs Tariff Classification with a Legal Argument | *Python, scikit-learn, LightGBM, PyTorch, DeepSeek, FastAPI, PostgreSQL*

Classifies a description of goods into Nigeria's 10-digit tariff (the ECOWAS CET) or the US's, and argues for the code as a customs broker would: the Note that excluded a competitor, the General Rule that settled a close call, the precedents behind it. It says how likely the code is to be right and itemises the landed duty in naira. Built on 220,625 crawled U.S. Customs rulings. 277 tests.

- **76.1%** right at six digits on 1,737 rulings from 2023 on, which no stage trained on, against 55.2% for an LLM given the description alone and 47.8% for copying the nearest past ruling.
- Three measured stages: a beam search over the tariff tree with the Notes compiled as hard filters and a trained ensemble inside it (64.1%); a LambdaRank reranker with two cross-encoders trained on 605k pairs mined from the first stage's own confusions (69.1%); and DeepSeek choosing among the top five only where the system is unsure (76.1%, about \$0.0003 a product).
- Calibrated confidence out of fold (ECE **0.029**), so autonomy is a measured setting: the line set for 90% right auto-accepts **43.9%** of held-out products at **90.8%** right and routes the rest to review with their candidates.
- The LLM earns its place only as the last step: on its own it scores 9 points below the beam, and thinking mode, a 4× larger model and five prompt variants all measured within noise, so none shipped.

Yoruba Speech and Tone Comparison Engine | *Python, Montreal Forced Aligner, wav2vec2, Parselmouth, scikit-learn, FastAPI*

Pronunciation assessment for a Yoruba learning app: an eight-stage pipeline (corpus audit, G2P, forced alignment, F0 extraction, tone classification, vowel identity, comparison, report) that scores a learner's recording against a prompt and says which word and which vowel went wrong. 208 tests.

- No reliable pretrained Yoruba aligner exists, so trained the acoustic model from scratch on ~118 h, 193,309 utterances and 2,801 speakers, every line diacritised; 99.85% of utterances align, against a 15,734-entry generated dictionary.
- Vowel identity **93.1%** accuracy, **0.912** macro F1 on 575,125 vowels from 2,786 speaker-disjoint held-out speakers; tone **79.6%** against a 37.8% majority baseline; substitutions caught at 94.0% for 3.8% false alarms, omissions at 83.9% for 7.2%.
- Found the ceiling was the representation, not the corpus: swapping 39 hand-designed acoustic measurements for a mean-pooled wav2vec2 layer moved macro F1 **0.432** → **0.811** with nothing else changed, and the same lesson held for tone: eight times more training data moved a gradient-boosted tree 69.6% → 70.2%, while reading the whole utterance with a BiGRU moved it to 79.6%.
- Replaced Montreal Forced Aligner with a neural forced aligner: the same acoustic-model-plus-Viterbi architecture, with the GMM swapped for a 2 MB head over the project's wav2vec2 encoder, distilled from MFA's own alignments. 86.2% of boundaries land within 20 ms of MFA's, and an assessment went from 14.3 s to 0.44 s: 96% of that time was MFA's per-invocation corpus setup, which no configuration reduces.
- Verifies the learner actually said the prompt via speech-coverage thresholding, because forced alignment cannot fail: it returns an alignment for any audio and any text, so without the check a learner saying something else entirely scores 100%.
- Serves at **4.7× realtime on CPU**, so no GPU is needed in production; calibration scripts score through the production code path rather than rebuilding feature rows, after an earlier version silently measured a different path than the one that shipped.

SEC 10-K Financial Intelligence Agent Harness | *Python, DeepSeek, LangGraph, MCP, networkx, BM25, XBRL*

Extracts provenance-checked intelligence from SEC 10-K filings, then runs two agent implementations over the same tools and the same task set so a harness, not an opinion, decides which orchestration ships. Five layers, 12 evaluation tasks.

- Every extracted object carries verbatim source text and every numeric claim is cross-checked against SEC XBRL companyfacts: **100% provenance integrity** across all six hand-verified companies, with risk recall 71–100% and value accuracy 67–100%.
- Three retrieval tiers and no vector database: a JSON fact store, a self-contained BM25 over passages, and a networkx knowledge graph (1,442 nodes, 1,906 edges over eight filings). The pipeline already emits structured provenance-carrying objects, so approximate search over embeddings would have been strictly worse than exact lookup.
- A custom think-act-observe loop and a LangGraph implementation share one identical tool set; the runner scores accuracy, latency, cost and trace, with a result store, regression detector and prompt versioning; the winner is served over an MCP server.
- The harness caught what code review did not: a chunk-size A/B cut tokens **6.7×** (108,848 → 16,288 output tokens, 17.2 → 2.4 min) with recall flat and severity accuracy *improving* 53.8% → 88.5%; a retrieval tier that had been a no-op for its entire life because the parser stripped the blank lines it split on; and XBRL, the audited ground truth, silently returning the wrong fiscal year for 52/53-week filers, since only the filer's own stamp separates JNJ's FY2022 from FY2021.

TheLayman | *FastAPI, PostgreSQL, multi-provider LLM, Render*

A paper explainer and daily discovery feed: ingests an arXiv URL, DOI or PDF and teaches it at three depths, backed by whichever of five LLM providers the account has keys for. 160 tests, deployed.

- Balancing the feed by **domain rather than native category** was the fix that made multi-source work: a per-category cap left computing holding 17 of 20 slots, since cs.AI, cs.LG and cs.CV are three categories but one domain; capping by domain brought it to 8.

GLOBAL COMPETITIONS

3rd, Humyn.ai Flowcaster Challenge 2023 · **4th**, Predict Potential Spammers on Fiverr · **8th**, Digital Turbine Auction Bid Price Prediction

EDUCATION

Federal University of Agriculture, Abeokuta (FUNAAB), Nigeria | B.Eng Mechatronics Engineering · 2017 – 2023

WorldQuant University | MSc Financial Engineering · 2025 – 2026

United States Artificial Intelligence Institute (USAII) | Certified Artificial Intelligence Engineer (CAIE) · 2023

AFFILIATIONS

ML Collective · AI+ FUNAAB (Student Member) · Data Scientists Network · Google Developer Student Club